



E-CELL
NIT TRICHY

E-CELL

TECH DOMAIN

T A S K 2

Task 1: E-Commerce Platform Development

Objective:

Develop a modern E-Commerce Platform that enables users to browse products, place orders, and complete purchases through a seamless shopping experience. The platform should include a high-conversion landing page, a customer-facing storefront, and an administrative dashboard for managing business operation

Customer-Facing Application

- Modern, visually appealing design
- Clear value proposition and branding
- High-conversion hero section with call-to-action
- Featured products and promotional banners
- Customer testimonials and trust indicators

Storefront

- Product listing and categorization
- Product detail pages
- Search and filtering functionality
- Shopping cart management
- Responsive design across all devices

Checkout Experience (Bonus Points)

- Secure checkout flow
- Address management
- Order summary and confirmation
- Coupon and discount code application
- Simulated Payment integration
- Order tracking

Admin Dashboard

Product Management

- Add, edit, and remove products
- Manage categories and inventory
- Upload product images
- Stock availability tracking

Analytics Dashboard

- Sales overview
- Revenue metrics
- Top-selling products
- Customer activity insights
- Conversion statistics

Order Management

- View and manage customer orders
- Update order status
- Order history and tracking

Coupon Management (Bonus points)

- Generate discount coupons
- Set expiration dates .
- Percentage-based and fixed discounts

Banner Management (Bonus Points)

- Create promotional banners
- Schedule banner visibility
- Manage homepage campaigns

Backend Requirements

- Authentication and authorization
- Role-based access control (Admin/User)
- Product and inventory management
- Order processing workflows
- Database integration
- Secure data handling

Learning Objectives

- Full-stack application development
- Frontend and backend integration
- Database design and management
- Authentication and authorization
- State management
- API development and consumption
- E-commerce workflows and business logic
- Dashboard and analytics development
- Scalable application architecture

Suggested Tech Stack:

Category	Technologies
Backend	Nodejs + ExpressJs
Frontend	HTML, CSS, JavaScript
Frameworks	React / Next.js
Styling	Tailwind CSS / Bootstrap
Animations	Framer Motion or other animation libraries
Additional Tools	Clouinary / Vercel / Render

Deliverables:

- Source code repository with documentation
- Responsive design across devices
- Deployed website link
- README containing setup instructions
- AI prompts used to make the website

SUBMISSION GUIDLINES:

**Meet the deadlines, late submissions will
not be entertained**

For any clarifications reach out to:

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